

Invertix: Data-Center Siting & Power

AI is driving a wave of new data centers, and the binding constraint is increasingly **electricity**. Siting one is a multi-way trade-off between price, carbon, grid congestion and connectivity. Can you help someone reason about where to build and how to power it?

WORTH EXPLORING

- Recommend locations for a given data-center size and explain the trade-offs.
- Plan a supply mix of grid, PPA and on-site generation against cost and carbon.
- Overlay capacity, prices, carbon and congestion to reveal good vs bad sites.

Invertix Open Track: Satellite Data for Solar

Satellites capture irradiance, cloud cover, and atmospheric conditions across the globe — data that ground sensors can't match in coverage or scale.
Can you turn it into something operators actually use?

WORTH EXPLORING

- A generation forecast that beats persistence using satellite-derived irradiance.
- A nowcast for intraday trading or grid balancing decisions. reveal good vs bad sites.
- Any tool that takes raw satellite feeds and turns them into an operational output.



: Digital Twins of Solar Plants

Find hidden **underperformance** in 10 year real solar plant monitoring data using **ML**

Solar plants are historically mostly judged by the Performance Ratio (PR). There is a clear need for advanced, data-driven analytics on inverter level that go beyond PR. Can you make model-based approaches analyzing degradation trends and find out the financial impacts from quality issues, downtimes and grid curtailments — and prove both with numbers?

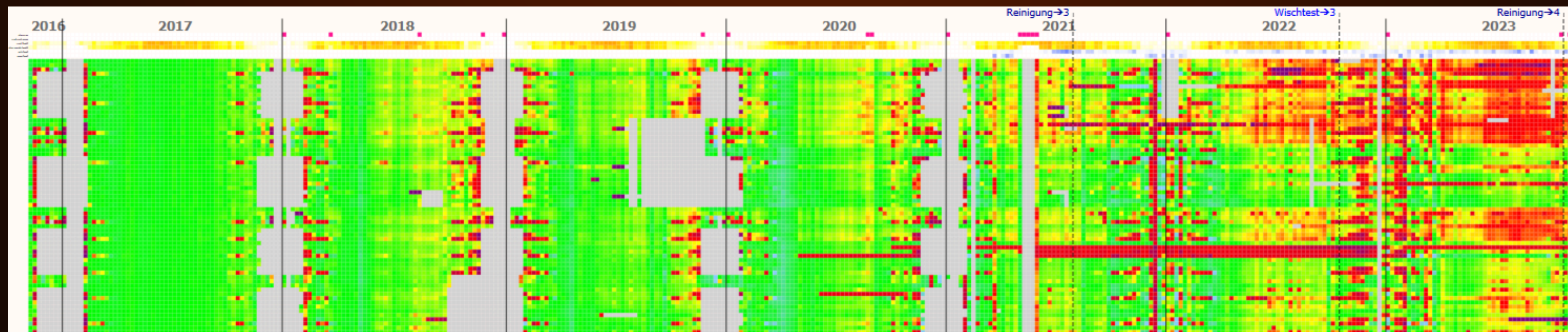
WORTH EXPLORING

- Build a per-inverter ML power model (optional also for current and voltage)
- Benchmark individual inverter models (trained with 1st year operation data) against real inverter dataset of 10 years!
Compare results for different module types over the whole timeline.



: Digital Twins of Solar Plants

Long Term Performance Tracking:



Open Track:

BUILT USEFULL AGENTS FOR THE PROVIDED DATASET

Build your own agents on top of 10 years of real solar plant monitoring and operational data from two utility-scale PV plants. Connect minute-resolution inverter telemetry, error codes, service tickets and plant design info — and turn it into something an O&M team would actually use.

WORTH EXPLORING

- Inverter failure detection
- Linking errorcode events to impact on inverter production
- Chatbot giving answers & visualisations of monitoring data
- Any agent that takes raw plant data and turns it into an O&M decision.

e-on: Grid Operation Agents

A grid must survive losing any single line or generator without anything else

overloading : staying "N-1 secure." When something trips, an operator has minutes to switch lines, redispatch generators or curtail output. Can an agent reason through that, propose safe low-cost actions, and explain them?

WORTH EXPLORING

- Take an overloaded grid back to a safe state, narrating its reasoning.
- Screen thousands of "what if this fails" cases; solve only the dangerous ones.
- Benchmark an LLM agent against a rule-based or optimization baseline.

pandapower Grid20p / L2RPN PyPSA-Eur real-time decisions

e-on Open Track: Grid Foundation Models

A pretrained grid model can predict powerflow dispatch **far faster than a solver**.
The interesting question is what near-instant prediction unlocks — interactive what-ifs, rapid screening, smarter optimizers. Can you build something useful on top of one, and stress-test where it holds up?

WORTH EXPLORING

- A “drag a load, see the grid respond” explorer powered by a fast model. Rank risky failure cases, then verify the worst with a real solver.
- An honest evaluation: how accurate is it, and where does it break?
- Anything that wouldn't be possible if inference took 10 seconds instead of 10 milliseconds.

GridSFM_Open pandapower gridfm-graphkit OPFData